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Step 1: Create a private Docker registry

```
~/desktop  ~zsh...  ~/desktop  ..esktop/my-app

base sateeshsahu@Sateeshs-MacBook-Air ~/desktop docker run -d -p 5000:5000 --restart=always --name private-registry registry:2
Unable to find image 'registry:2' locally
2: Pulling from library/registry
ddcb6d98388d: Pull complete
9a6b3f59ebc2: Pull complete
bd39ca3613a6: Pull complete
8bbaad1488a8: Pull complete
95459497489f: Pull complete
c4472e68032e: Download complete
94a5ce400b19: Download complete
Digest: sha256:a3d8aaa63ed8681a604f1dea0aa03f100d5895b6a58ace528858a7b332415373
Status: Downloaded newer image for registry:2
716e0d621bcb46be14c0dd6c197363d1d61f134bfb0dffe0aaba338683c21309
```

Step 2: Build a Docker image

```
base sateeshsahu@Sateeshs-MacBook-Air ~/desktop mkdir my-app && cd my-app
base sateeshsahu@Sateeshs-MacBook-Air ~/desktop/my-app echo -e 'FROM alpine\nCMD ["echo", "Hello from the private registry!"]' > Dockerfile
base sateeshsahu@Sateeshs-MacBook-Air ~/desktop/my-app docker build -t test-app:v1 .
[+] Building 4.0s (6/6) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 98B
=> [internal] load metadata for docker.io/library/alpine:latest
=> [auth] library/alpine:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 28B
=> [1/1] FROM docker.io/library/alpine:latest@sha256:5b10f432ef3da1b8d4c7eb6c487f2f5a8f096bc91145e68878dd4a5019afde11
=> resolve docker.io/library/alpine:latest@sha256:5b10f432ef3da1b8d4c7eb6c487f2f5a8f096bc91145e68878dd4a5019afde11
=> exporting image
=> => exporting layers
=> => exporting manifest sha256:dfcad4079e56ab8f4797972c8859e2518e5f980c42d678d88079d5548fa37254
=> => exporting config sha256:1a1e9562905ac1c323c28051eebe45b105b8c2adac23ae0cd680956d136b101d
=> => exporting attestation manifest sha256:ccc7b093c00e77c5ccaf91ce56ce292e2a1a2fd181ec848cd64ed85c3b8e8071
=> => exporting manifest list sha256:cc3d48e19f2beab39ce6ed95cc415237f96d049c9f95c0968ca0ddf39fe4cd7f
=> naming to docker.io/library/test-app:v1
=> unpacking to docker.io/library/test-app:v1

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/nh5vu236q9pvonh88omumxyvh
```

Step 3: Tag and push the image to the private registry

```
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/nh5vu236q9pvonh88omumxyvh
base sateeshsahu@Sateeshs-MacBook-Air ~/desktop/my-app docker tag test-app:v1 localhost:5000/test-app:v1
base sateeshsahu@Sateeshs-MacBook-Air ~/desktop/my-app docker push localhost:5000/test-app:v1
The push refers to repository [localhost:5000/test-app]
d17f077ada11: Pushed
cdc29a9dd3b9: Pushed
v1: digest: sha256:cc3d48e19f2beab39ce6ed95cc415237f96d049c9f95c0968ca0ddf39fe4cd7f size: 855
```

Step 4: Verify the registry using curl API

```
base sateeshsahu@Sateeshs-MacBook-Air ~/desktop/my-app curl -X GET http://localhost:5000/v2/_catalog
{"repositories":["test-app"]}
base sateeshsahu@Sateeshs-MacBook-Air ~/desktop/my-app curl -X GET http://localhost:5000/v2/test-app/tags/list
{"name":"test-app","tags":["v1"]}
```

Step 5: Run a container using host network mode

```
base sateeshsahu@Sateeshs-MacBook-Air ~/desktop/my-app docker run -d --network host --name my-host-nginx nginx:alpine
Unable to find image 'nginx:alpine' locally
alpine: Pulling from library/nginx
Digest: sha256:5616878291a2eed594aee8db4dade5878cf7edcb475e59193904b198d9b830de
Status: Downloaded newer image for nginx:alpine
fdd54113291f3d8b858cdf5bbb597493418e58e1efc93c66d9ee59e0b794d6e
```

Step 6: Explain the difference between host and bridge networking

Bridge Networking (Default): The container gets its own isolated network namespace and its own internal IP address on a virtual bridge (usually docker0). To allow outside traffic to reach the container, you must explicitly map ports using the -p flag (-p 8080:80). It provides excellent isolation but introduces a slight network translation overhead.

Host Networking (--network host): The container shares the host machine's network namespace completely. The container does not get its own IP address; instead, it uses the host's IP. If a container runs an app on port 80, it binds directly to port 80 of the physical host machine. It offers higher performance because there is no network translation, but it provides zero network isolation and can lead to port conflicts.

Step 7: Pull the image from the private registry

```
base [2] sateeshsahu@Sateeshs-MacBook-Air [2] ~/desktop/my-app [2] docker rmi localhost:5000/test-app:v1
Untagged: localhost:5000/test-app:v1
base [2] sateeshsahu@Sateeshs-MacBook-Air [2] ~/desktop/my-app [2] docker pull localhost:5000/test-app:v1
v1: Pulling from test-app
Digest: sha256:cc3d48e19f2beab39ce6ed95cc415237f96d049c9f95c0968ca0ddf39fe4cd7f
Status: Downloaded newer image for localhost:5000/test-app:v1
localhost:5000/test-app:v1
base [2] sateeshsahu@Sateeshs-MacBook-Air [2] ~/desktop/my-app [2] docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
custom-nginx:v1	6905a191cc38	91.8MB	25.8MB	
httpd:latest	89a0b59e6d72	207MB	47.8MB	U
localhost:5000/test-app:v1	cc3d48e19f2b	13.6MB	4.2MB	
mongo:latest	7abfba0d07c9	1.27GB	325MB	U
mysql:latest	c9e48b0c008f	1.32GB	287MB	U
nginx:alpine	5616878291a2	92.7MB	26.7MB	U
nginx:latest	6e23479198b9	258MB	64.2MB	U
node:latest	3953ec6a2c10	1.63GB	415MB	U
postgres:15-alpine	1c52f5ad23db	386MB	108MB	U
redis:7-alpine	7aec734b2bb2	61.9MB	18.2MB	U
registry:2	a3d8aaa63ed8	36.3MB	9.67MB	U
snap-pdf:latest	82975dd08c39	5.56GB	1.43GB	U
test-app:v1	cc3d48e19f2b	13.6MB	4.2MB	
two-tier-backend:latest	54566ea5376d	583MB	139MB	U

Step 8: Explain the concept of authentication in Docker registry

By default, the open-source Docker Registry allows anonymous access—anyone who can reach the network port can push or pull images. In a production environment, this is a massive security risk. Authentication restricts access so that only authorized users or systems can interact with it.